

9b

2. Analog Filter Design

The transfer function $H(s)$ of a low pass Butterworth Filter can be expressed in terms of the product:

$$H(s)H(-s) = \frac{1}{1 + \left(\frac{s}{j\omega_c}\right)^{2N}}$$

where N is the order of the filter and ω_c is the filter cutoff frequency.

The poles of $H(s)H(-s)$ are found from

$$s_p = (-1)^{\frac{1}{2N}} \cdot j\omega_c = \omega_c e^{j\frac{(2k+1+N)\pi}{2N}}, \quad k \text{ is an integer}$$

The $2N$ poles occur in pairs on a circle of radius ω_c in the s -plane with angular spacing of $\frac{\pi}{N}$ radians. The pole pairs are symmetrical about the $j\omega$ axis. To construct a stable causal filter, we choose the N poles along the left half semicircle in the s -plane for $H(s)$. The remaining N poles in the mirrored right half plane are poles from $H(-s)$.

- a) Derive the transfer function $H(s)$ of a 4th order Butterworth Filter. You may express your answer as the product of two second-order systems in canonical form. Let the passband edge frequency $f_p = 10\text{Hz}$. The passband edge frequency f_p (in Hz) is defined as the frequency when $|H(j\omega)|^2$ or $|H(j2\pi f)|^2$ drops to $\frac{1}{2}$ of its value at zero frequency.

Plot the Bode plots of the corresponding FRF $H(j\omega)$ of the 4th order Butterworth filter.

- Submit both the derivation and plot.

We can use the equation for $|H(s)|$, with

$$\omega_p = 2\pi f = 20\pi.$$

$$|H(s)_4| = \frac{1}{1 + \left(\frac{s}{j\omega_c}\right)^8}$$

With poles spaced evenly at radius ω_c at $\frac{\pi}{4}$ on the left half of the plane. Following the equation, we have poles at

$$\frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$$

Which is equivalent to

$$-\frac{5\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, -\frac{7\pi}{8}$$

We're only looking at the left half plane, which has poles at

$$H(s) = \frac{\omega_c^4}{(s - \omega_c e^{j\frac{5\pi}{8}})(s - \omega_c e^{j\frac{7\pi}{8}})(s - \omega_c e^{j\frac{9\pi}{8}})(s - \omega_c e^{j\frac{11\pi}{8}})}$$

Which simplifies down to

$$H(s) = \frac{\omega_c^2}{(s - \omega_c e^{j\frac{5\pi}{8}})(s - \omega_c e^{j\frac{7\pi}{8}})} \frac{\omega_c^2}{(s - \omega_c e^{j\frac{9\pi}{8}})(s - \omega_c e^{j\frac{11\pi}{8}})}$$

We can simplify each of these by expanding to get

$$H(s) = \left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{\frac{j5\pi}{8}} + e^{\frac{-j5\pi}{8}}) + \omega_c^2} \right) \left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{\frac{j7\pi}{8}} + e^{\frac{-j7\pi}{8}}) + \omega_c^2} \right)$$

We can use the given equation for the magnitude

$|H(s)| = H(s)H(-s)$ to set find ω_p

$$\frac{|H(j\omega_p)|}{H(0)} = \frac{1}{\sqrt{2}}$$

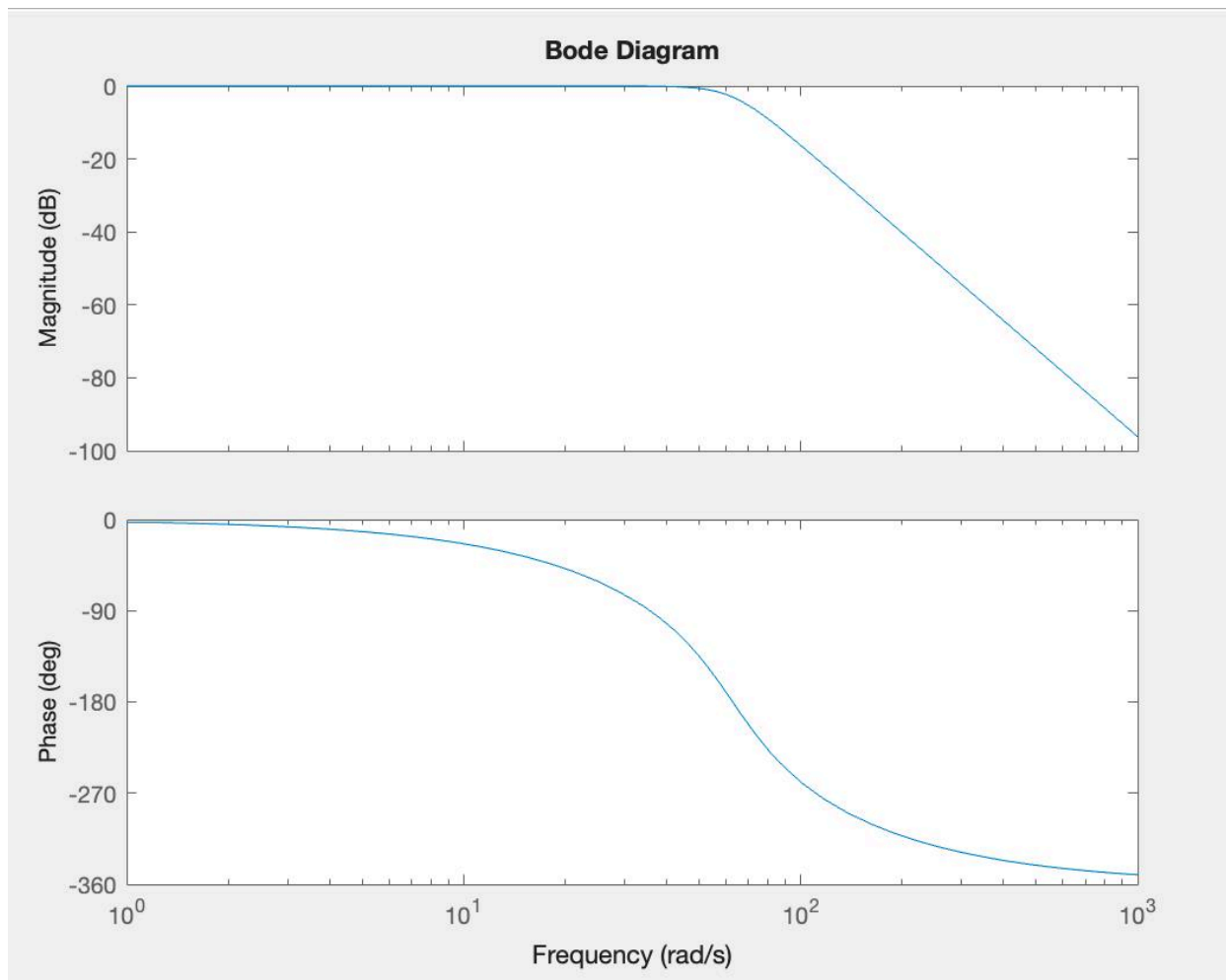
We can make this far simpler by considering

$$H(j\omega_p)H(-j\omega_p) = \frac{1}{2}$$

$$\frac{1}{1 + \left(\frac{j\omega_p}{j\omega_c}\right)^8} = \frac{1}{2}$$

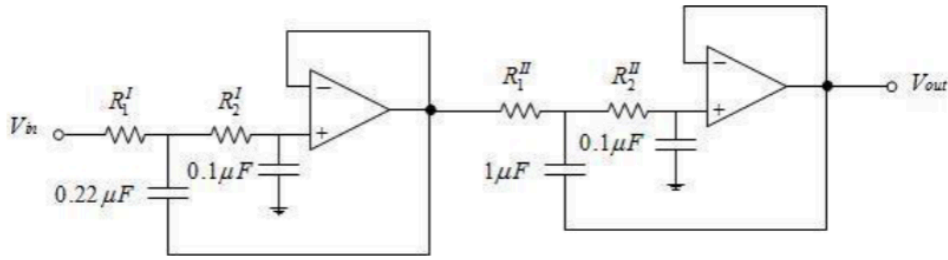
This is true when the numerator is $1 + \left(\frac{\omega_p}{\omega_c}\right)^8 = 2$

so $\omega_c = \omega_p = 20\pi$.



Note that the corner is at $20 * \pi$, 62.8.

- b) The 4th order Butterworth Filter above is implemented with a two stage Sallen-Key circuit:



In each stage, Sallen-Key circuit transfer function $H(s)$ is given by the following equation:

$$H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + s(R_1 + R_2)C_2 + 1}$$

Given the capacitor values shown in the diagram, i.e. we have chosen $C_1^I = 0.22\mu F$, $C_2^I = 0.1\mu F$, $C_1^{II} = 1\mu F$, $C_2^{II} = 0.1\mu F$.

Select “Standard” resistors $R_1^I, R_2^I, R_1^{II}, R_2^{II}$ to design the filter for $f_p \approx 10\text{Hz}$. Standard resistors come in 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2; multiplied by decade values (+/- 10%). **The solution should result in a set of more damped (higher ξ) poles in the first stage, and a less damped (lower ξ) poles in the second stage.**

- Submit the calculation of the design values $R_1^I, R_2^I, R_1^{II}, R_2^{II}$. Put the final design values in a table (calculated vs. closed standard resistors/capacitors).

We found before two second order functions. We will make one for each circuit.

$$H(s) = \left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{j\frac{5\pi}{8}} + e^{-j\frac{5\pi}{8}}) + \omega_c^2} \right) \left(\frac{\omega_c^2}{s^2 - 2\omega_c s(e^{j\frac{7\pi}{8}} + e^{-j\frac{7\pi}{8}}) + \omega_c^2} \right)$$

We can rewrite this into a nice form for each, simplifying $e^{j\phi} + e^{-j\phi}$ to $2\cos(\phi)$.

$$H(s)_{\text{part}} = \frac{1}{\frac{s^2}{\omega_n^2} - \frac{4\cos\phi}{\omega_n} + 1}$$

Pattern matching, we have

$$R_1 R_2 C_1 C_2 = \left(\frac{1}{\omega_n} \right)^2$$

and

$$(R_1 + R_2)C_2 = 4 \frac{\cos\phi}{\omega_n}$$

So we have

$$R_1 R_2 = \frac{1}{C_1 C_2} \frac{1}{\omega_n^2}$$

$$R_1 + R_2 = \frac{4}{C_2} \frac{1}{\omega_n} \cos\phi$$

We want to turn each into the form

$$H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + s(R_1 + R_2)C_2 + 1}$$

we have the generic form

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n + \omega_n^2} = \frac{1}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}$$

Lets make this more concrete by tying to our system $H(s)$.

Where $\phi = \frac{5\pi}{8}$ or $\frac{7\pi}{8}$.

This tells us that

$$\frac{1}{\omega_n^2} = R_1 R_2 C_1 C_2$$

and that

$$(R_1 + R_2)C_2 = \frac{2\zeta}{\omega_n}$$

We will have a system of equations with constants that has solutions with

$$\begin{aligned} R_1 + R_2 &= C \\ R_1 R_2 &= D \\ R_2 &= C - R_1 \\ R_1(C - R_1) &= D \\ R_1 C - R_1^2 &= D \\ R_1^2 - R_1 C + D &= 0 \\ R_1 &= \frac{C \pm \sqrt{C^2 - 4D}}{2} \\ R_2 &= \frac{D}{R_1} \end{aligned}$$

							Standard resistor values	
w_c	6.283E+01	- Cos					Coefficients	Decades
phi 1	-1.963E+00	3.827E-01	H_1(j w_c)	2.761E-01			1.000E+00	1.000E+00
phi 2	-2.749E+00	9.239E-01					1.200E+00	1.000E+01
c11	2.200E-07		H_2(j w_c)	1.790E-01			1.500E+00	1.000E+02
c12	1.000E-07		H_tot(j w_c)	4.942E-02			1.800E+00	1.000E+03
c21	1.000E-06						2.200E+00	1.000E+04
c22	1.000E-07						2.700E+00	1.000E+05
r11	1.800E+05						3.300E+00	1.000E+06
r12	6.800E+04						3.900E+00	1.000E+07
r21	5.600E+05						4.700E+00	
r22	4.700E+03						5.600E+00	
							6.800E+00	
							8.200E+00	
Set 1		Actual	Goal	10% tolerance	Power of Ten	Diff		
zeta	(R1+R2)C2	2.480E-02	2.436E-02	2.436E-03	-1.613E+00	-4.376E-04		
w_n	sqrt(1/R1R2C1C2)	6.094E+01	6.283E+01	6.283E+00	1.798E+00	1.892E+00		
C	R1+R2	2.480E+05	2.436E+05	2.436E+04	5.387E+00	-4.376E+03		
D	R1R2	1.224E+10	1.151E+10	1.151E+09	1.006E+01	-7.262E+08		

	R1	1.800E+05	1.795E+05	1.795E+04	5.254E+00	-5.307E+02		
	R2	6.800E+04	6.415E+04	6.415E+03	4.807E+00	-3.845E+03		
Set 2		Actual	Goal	10% tolerance	Power of Ten			
zeta	(R1+R2)C2	5.647E-02	5.882E-02	5.882E-03	-1.231E+00	2.346E-03		
w_n	sqrt(1/R1R2C1C2)	6.164E+01	6.283E+01	6.283E+00	1.798E+00	1.193E+00		
C	R1+R2	5.647E+05	5.882E+05	5.882E+04	5.769E+00	2.346E+04		
D	R1R2	2.632E+09	2.533E+09	2.533E+08	9.404E+00	-9.897E+07		
	R1	5.600E+05	5.838E+05	5.838E+04	5.766E+00	2.382E+04		
	R2	4.700E+03	4.339E+03	4.339E+02	3.637E+00	-3.613E+02		

With the components for the first and second circuit given by

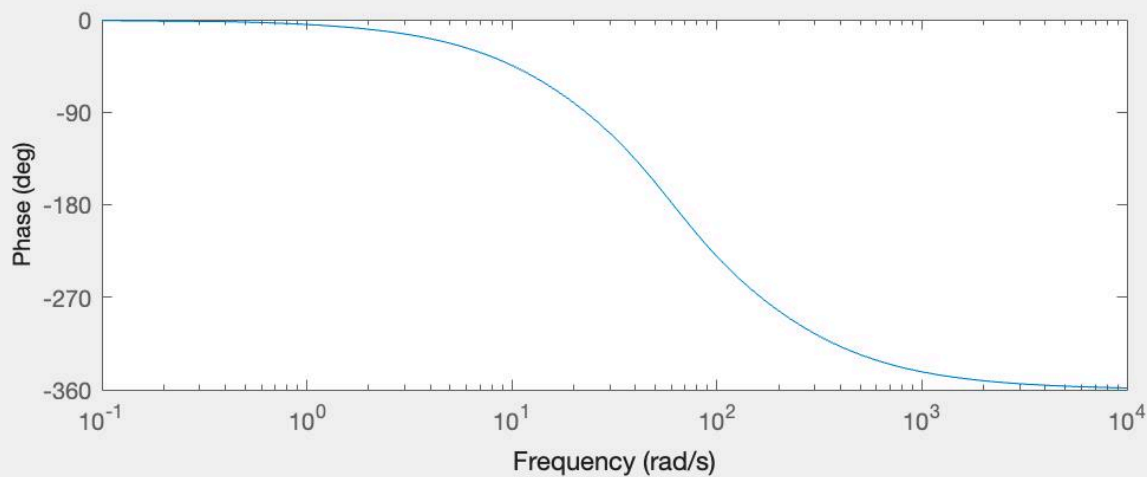
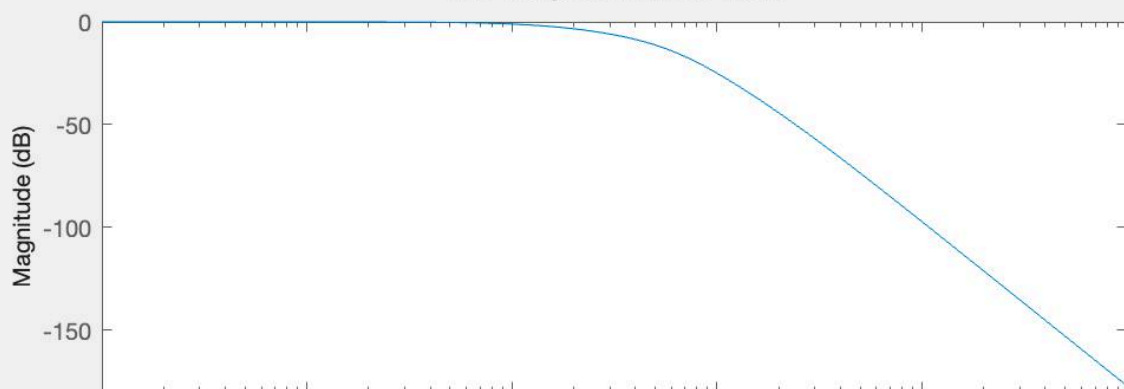
c11	2.200E-07	F
c12	1.000E-07	F
c21	1.000E-06	F
c22	1.000E-07	F
r11	1.800E+05	Ω
r12	6.800E+04	Ω
r21	5.600E+05	Ω
r22	4.700E+03	Ω

- c) Create a Bode plot of the overall circuit frequency response function $|H(j\omega)|$ with the selected standard component values. Determine the passband edge frequency f_p (in Hz) from the plot.

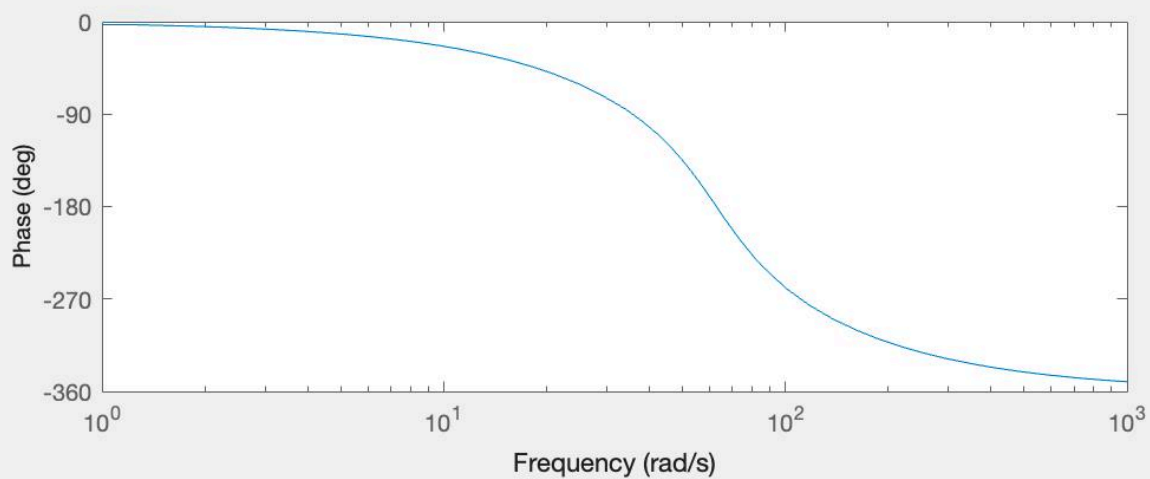
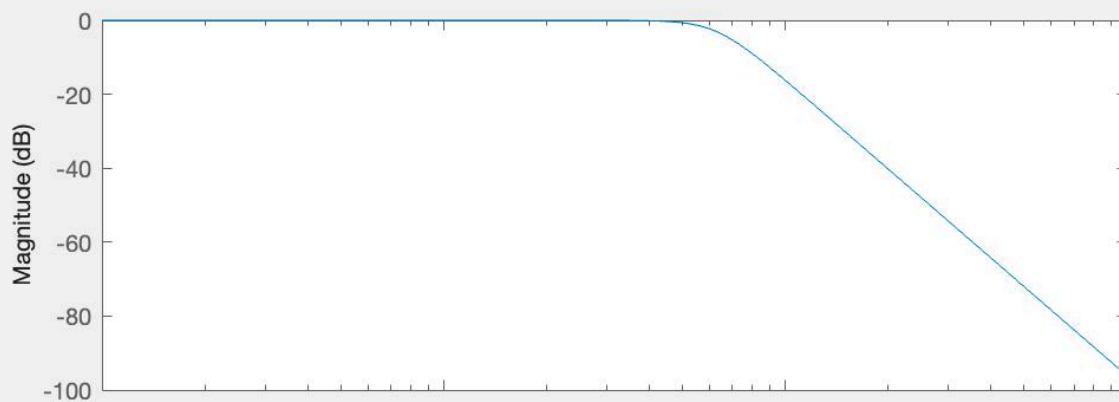
- Submit the labeled plot

Note: Your final experimental kit will have slightly different R and C values, so you should code it up for easy adjustment and evaluate the discrepancy involved in the final implementation.

Two-Stage Sallen Key Filter



Optimal Butterworth



3. Digital Filter Design

A comb filter is a causal digital filter with zeros evenly spaced around the unit circle in the z-plane.

$$H_{comb}(z) = b_0(1 - z^{-M})$$

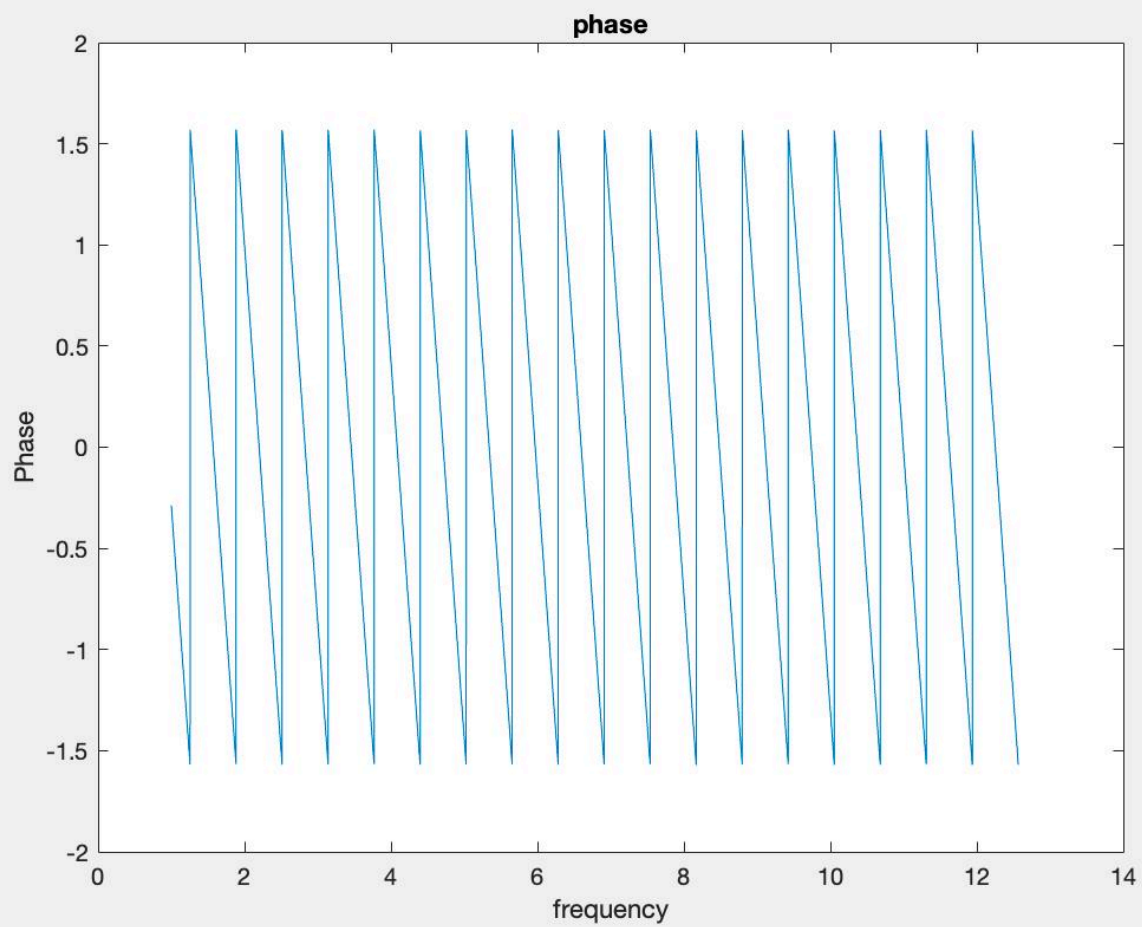
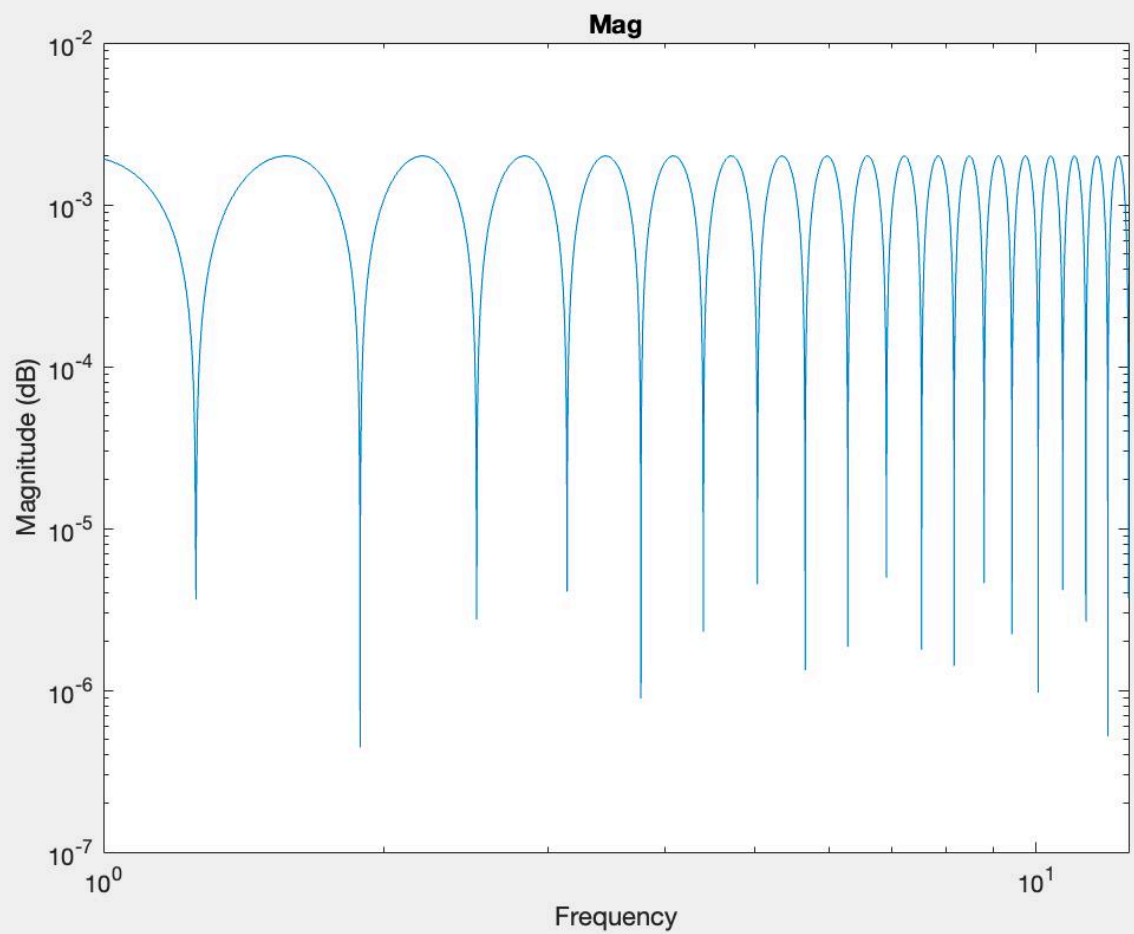
where M is the number of zeros. By selectively removing conjugate pairs of zeros, the designer can provide passbands at the selected frequencies with stopbands at the frequencies of the remaining zeros.

- a) Plot the magnitude of the frequency response function $|H_{comb}(e^{j\hat{\omega}})|$ of an M=10 comb filter

- Submit the labeled plot.

```

                                b0 = 1
                                m = 10
Hcomb_func = @(z)b0 * (1 - z-M)
z = i * (1 : 10 : 10e4)
fori = 1 : length(z)
Hcomb(i) = Hcomb_func(z(i))
                                end
                                plus plots
```



b) Remove the three zeros located at $z = e^{j0}, e^{\pm j\frac{\pi}{5}}$. This allows frequencies at $\hat{\omega} = \pm \frac{\pi}{5}$ to pass through without being zeroed out. Plot the magnitude of the normalized frequency response $|H_{modified_comb}(e^{j\hat{\omega}})|$ of the modified comb filter. For normalized frequency response, **choose b_0 so that $|H_{modified_comb}(e^{j0})|=1$** . You may find conv() in MATLAB a helpful tool to expand the product of polynomials.

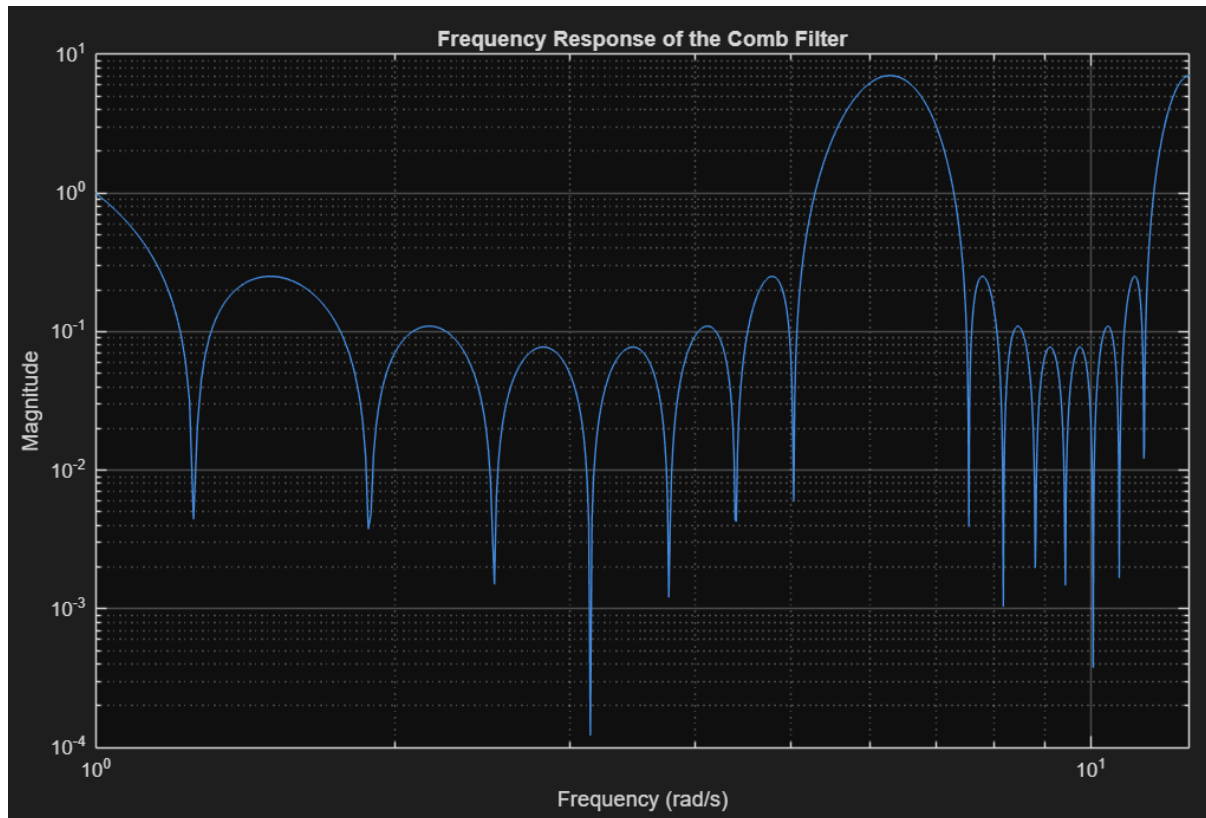
- Submit the labeled plot and identify the value of $|H_{modified_comb}(e^{j\frac{\pi}{5}})|$.
- In the final project, the Arduino uses a sampling interval of $T_s=10\text{ms}$ (see part 1 of HW 9). What is the corresponding continuous time frequency f in Hz that corresponds to a digital frequency of $\hat{\omega} = \pm \frac{\pi}{5}$?

Remove zeros from $e^{\pm j\frac{\pi}{5}}$ and e^0 .

We rewrite but divide by the poles to get

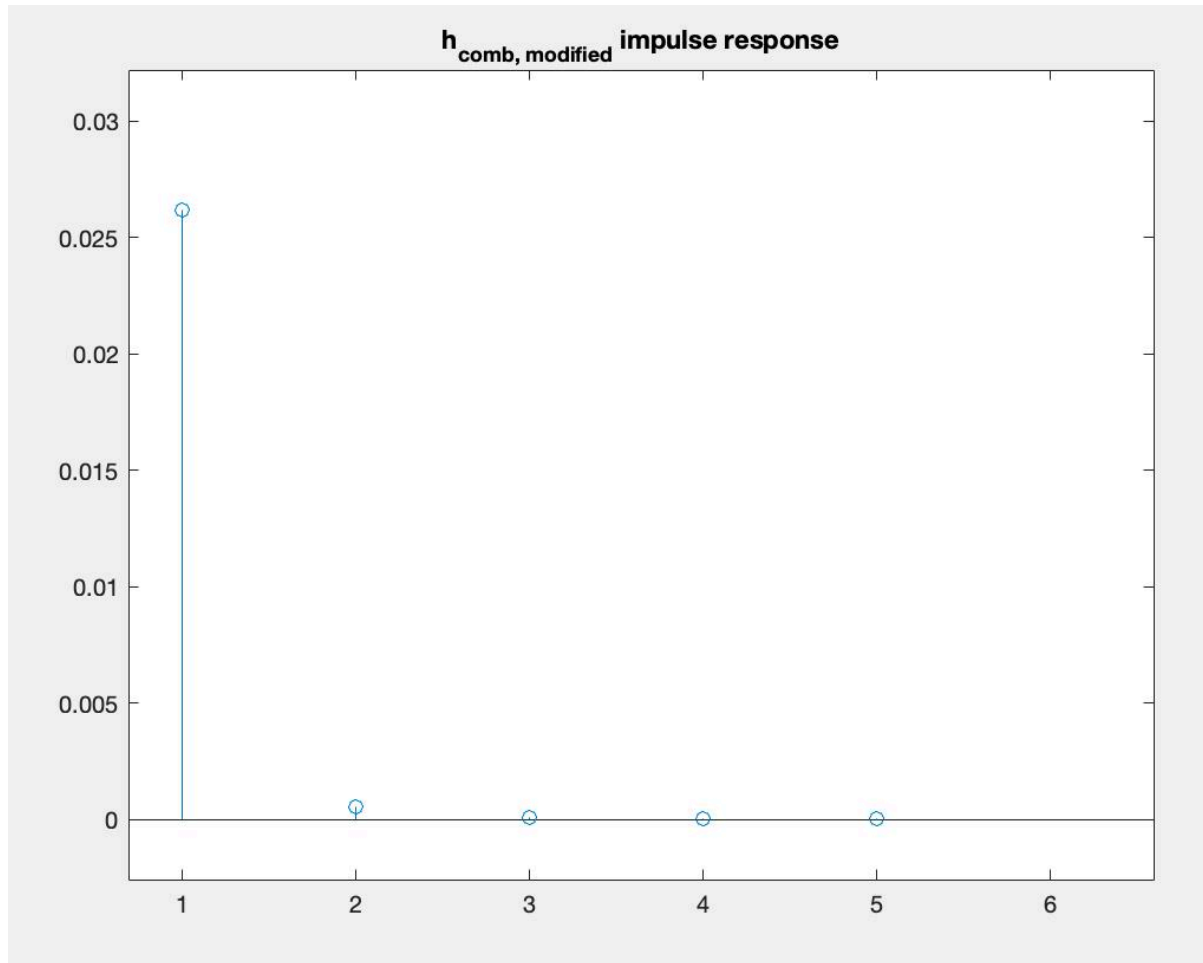
$$\frac{b_0(1 - z^{-M})}{(z - e^{j0})(z - e^{j\frac{\pi}{5}})(z - e^{-j\frac{\pi}{5}})}$$

But setting b_0 to be $\frac{1}{3.7217}$ (which is the amplitude at 0 if b_0 were 1).



c) Plot the impulse response of the filter $h_{modified_comb}[n]$. List the corresponding filter coefficients in a table.

- Submit the labeled plot.



h_{comb, modified} impulse response

